Tavonga Obey Chitambira

Full Stack Engineer

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Available upon graduation (March 2028)
Zimbabwean

Summary

Full-stack engineer and Computer Science student at Vishwakarma University, Pune. I build real-time distributed systems and data-driven web applications — currently architecting SyncBoard, a horizontally-scaled multiplayer incident management platform. My core stack is Next.js, TypeScript, Apollo GraphQL, Prisma, and PostgreSQL, with a growing focus on GenAI and LLM engineering.

Skills

Technical Profiency ●●●●○

Languages: TypeScript, JavaScript, Python, Java, SQL
Frontend: Next.js App Router, Tailwind CSS, shadcn/ui, Zustand, Apollo Client
Backend: Node.js, Express, Apollo GraphQL, Prisma, Socket.io, BullMQ
Databases & Infrastructure: PostgreSQL, Redis, Docker, Kubernetes (HPA)
Tools & Platforms: Cloudinary, Stripe, Git, Vercel

Languages

English — Fluent
Shona — Native

Education

BSc (Hons) in Computer Science,
Vishwakarma University, Pune
08/2024 – Present | India
Relevant coursework: Discrete Mathematics, Data Structures & Algorithms, Object-Oriented Programming and Database Systems

Projects

SyncBoard
07/2026 – Present
Real-Time Incident Management Platform *Next.js · Socket.io · Redis · PostgreSQL · Yjs · Zustand · BullMQ · Stripe · Kubernetes*

- Architected a horizontally-scaled WebSocket cluster using @socket.io/redis-adapter for Redis Pub/Sub fan-out across N pods with zero message loss
- Implemented Yjs CRDT for conflict-free collaborative text editing and optimistic locking with a version column for structured field conflict resolution (409 rollback on stale writes)
- Built a Redis Sorted Set presence system with 30s heartbeat, 45s TTL window, and background cleanup — surfacing live collaborators per incident room
- Designed a debounced PostgreSQL flush worker (500ms interval) to protect the database from raw socket event volume
- Wired Stripe seat licensing and per-incident usage metering via BullMQ async jobs, fully decoupled from the socket hot path
- Implemented reconnect delta sync with a 15-minute operation replay window and per-connection Redis sliding-window rate limiting (100 events/60s)

Courses

07/2026 – Present
GenAI & LLM Engineering — Self-directed, 8-phase structured roadmap *Currently completing: Python, ML Math foundations (NumPy, linear algebra, MML — Deisenroth)*

TCS Data Science Industry Project *Retail sales performance analysis: ARIMA forecasting, K-Means segmentation, Random Forest & XGBoost prediction, CLV estimation on AdventureWorks dataset*